CROP YIELD PREDICTION

Using Regression in Tamil Nadu

A simple data-based look at paddy yield trends and future estimates

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Why look at crop yield?

The idea is to use past records to make a sensible estimate for the next season.

- Agriculture depends on rainfall, water availability, soil, and farming practices.
- Yield can rise or fall from one year to another, so planning is not always easy.
- Regression helps us study the general trend in historical yield data.
- The estimate can support planning for storage, transport, and farm resources.

Focus crop

Paddy (rice)

A major food crop in Tamil Nadu, with yield affected by both weather and farm inputs.

Aim and objectives

What this small project tries to do

- Aim: estimate Tamil Nadu paddy yield using a simple linear regression model.
- Study the year-wise pattern in yield instead of relying on one year's result.
- Draw a best-fit line to show the overall direction of the data.
- Compare the model's fitted values with the example historical values.
- Discuss where regression is useful and where it can go wrong.

Data and variables

For a real submission, replace the demo values with official year-wise data.

Independent variable (X)

Year (2015–2024)

The year is used as the input to capture the long-term trend.

Dependent variable (Y)

Paddy yield (kg/ha)

Yield is the value the model tries to estimate from the year.

Important: these values are illustrative for demonstrating the method, not official measured statistics.

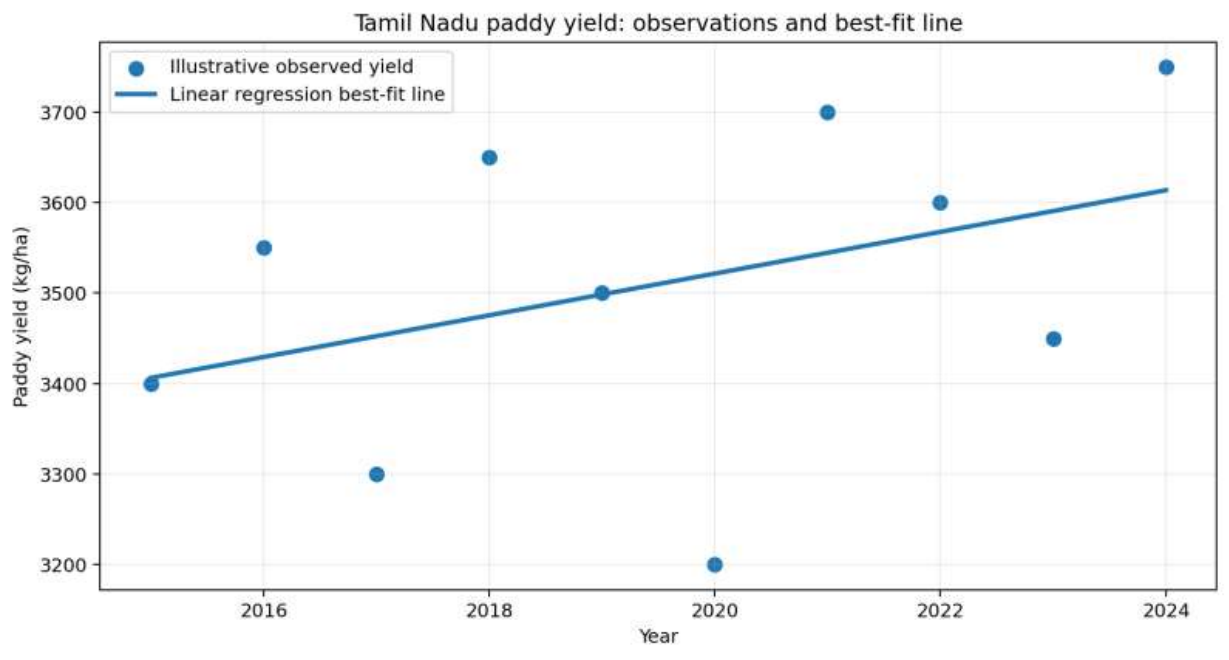
How linear regression works

It fits a straight line that stays as close as possible to the observed points.

Best-fit line equation

$$\hat{y} = a + bX$$

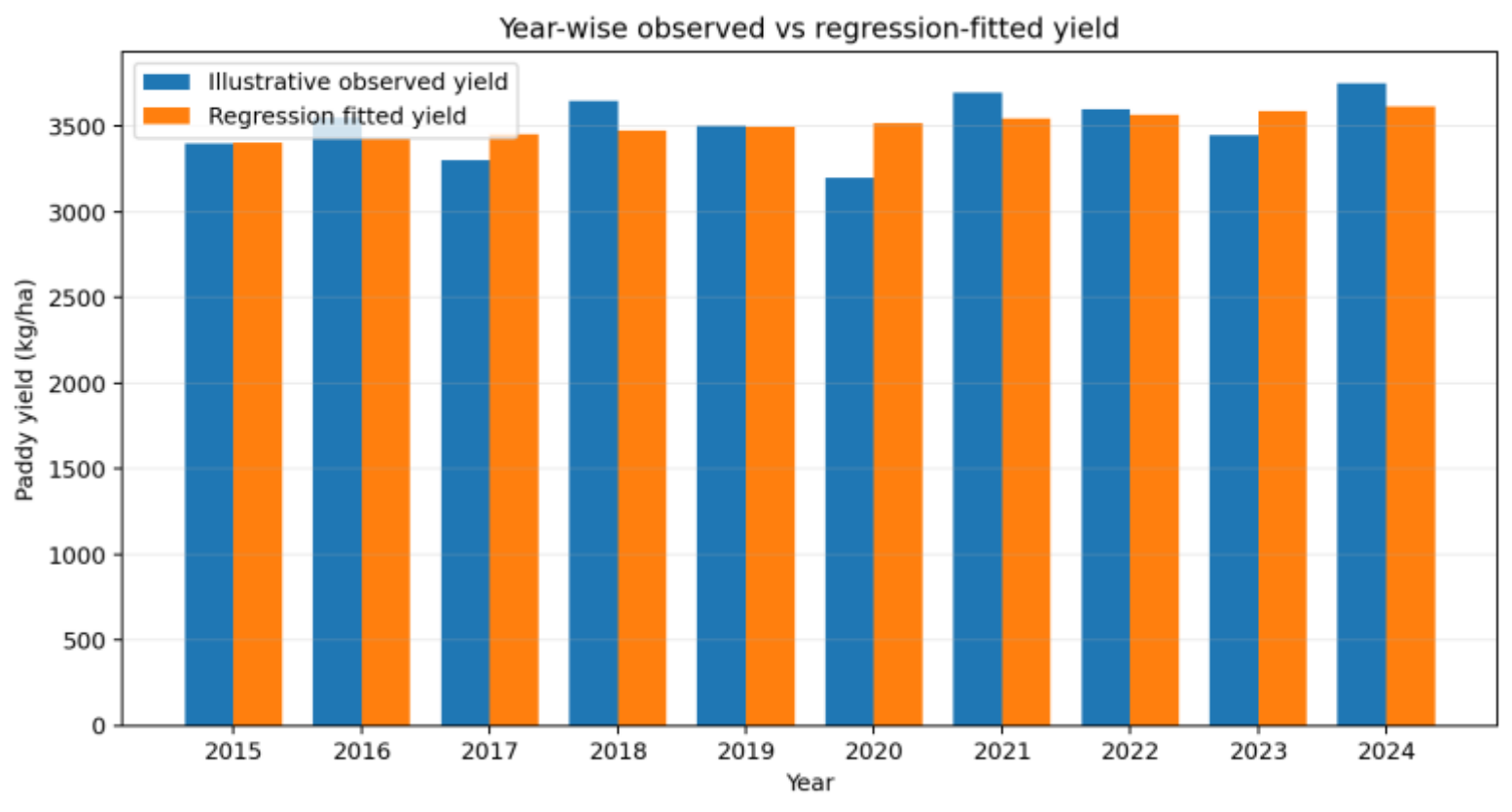
- X = year
- $\hat{y}$ = predicted yield
- a = intercept; b = slope



Illustrative fit: slope ≈ 23.0 kg/ha per year; R^2 ≈ 0.16

Year-wise comparison

Observed and fitted values are compared to see how closely the line follows the data.



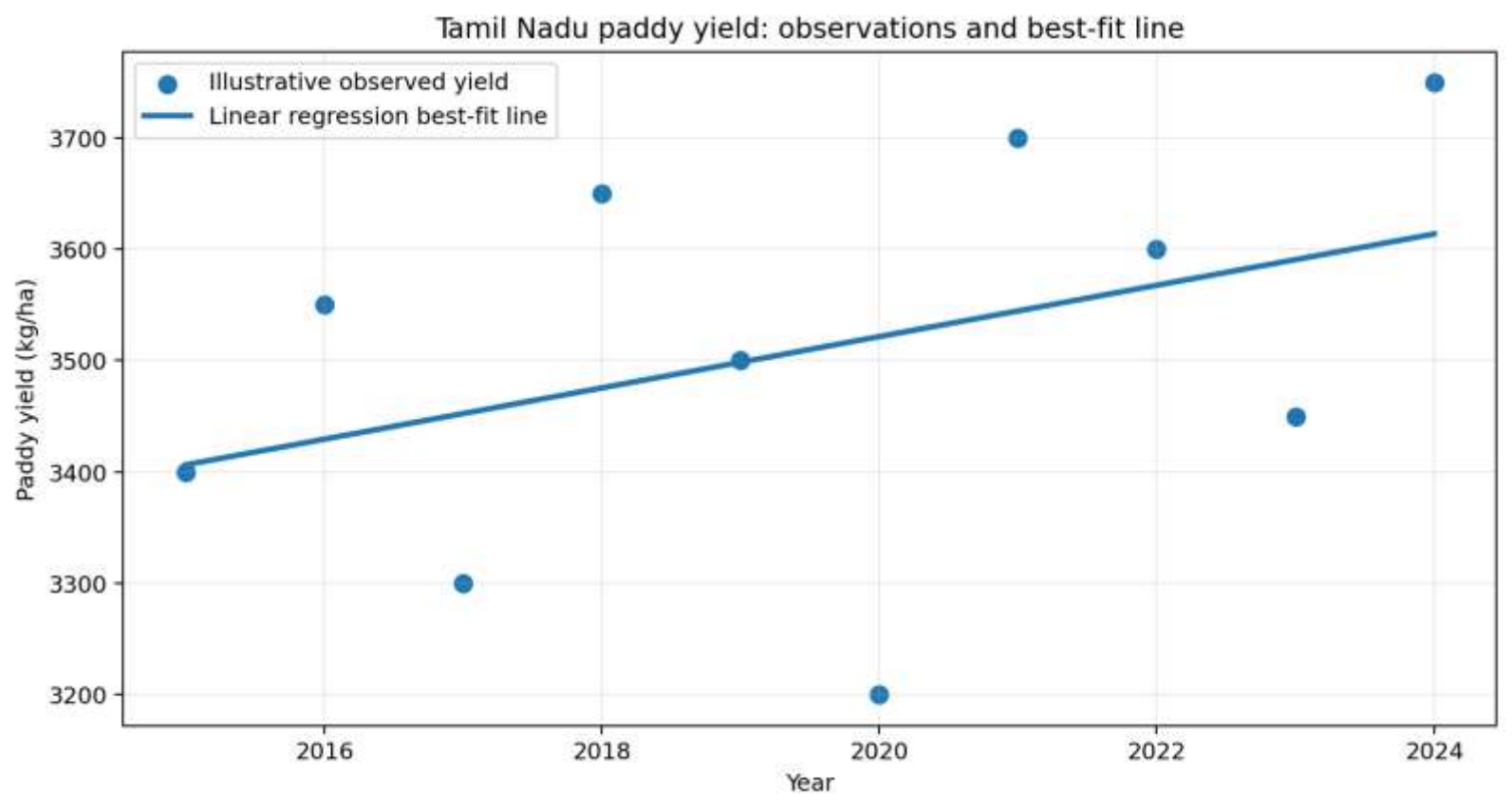
Example values (kg/ha)

Year	Observed	Fitted
2015	3400	3406
2016	3550	3429
2017	3300	3452
2018	3650	3475
2019	3500	3498
2020	3200	3522
2021	3700	3545
2022	3600	3568
2023	3450	3591
2024	3750	3614

Fitted = line's estimate for that year.

The best-fit line

A straight line summarises the overall trend; it does not pass through every point.



Regression equation

$$\hat{y} = -89060.3 + (23.0 \times \text{Year})$$

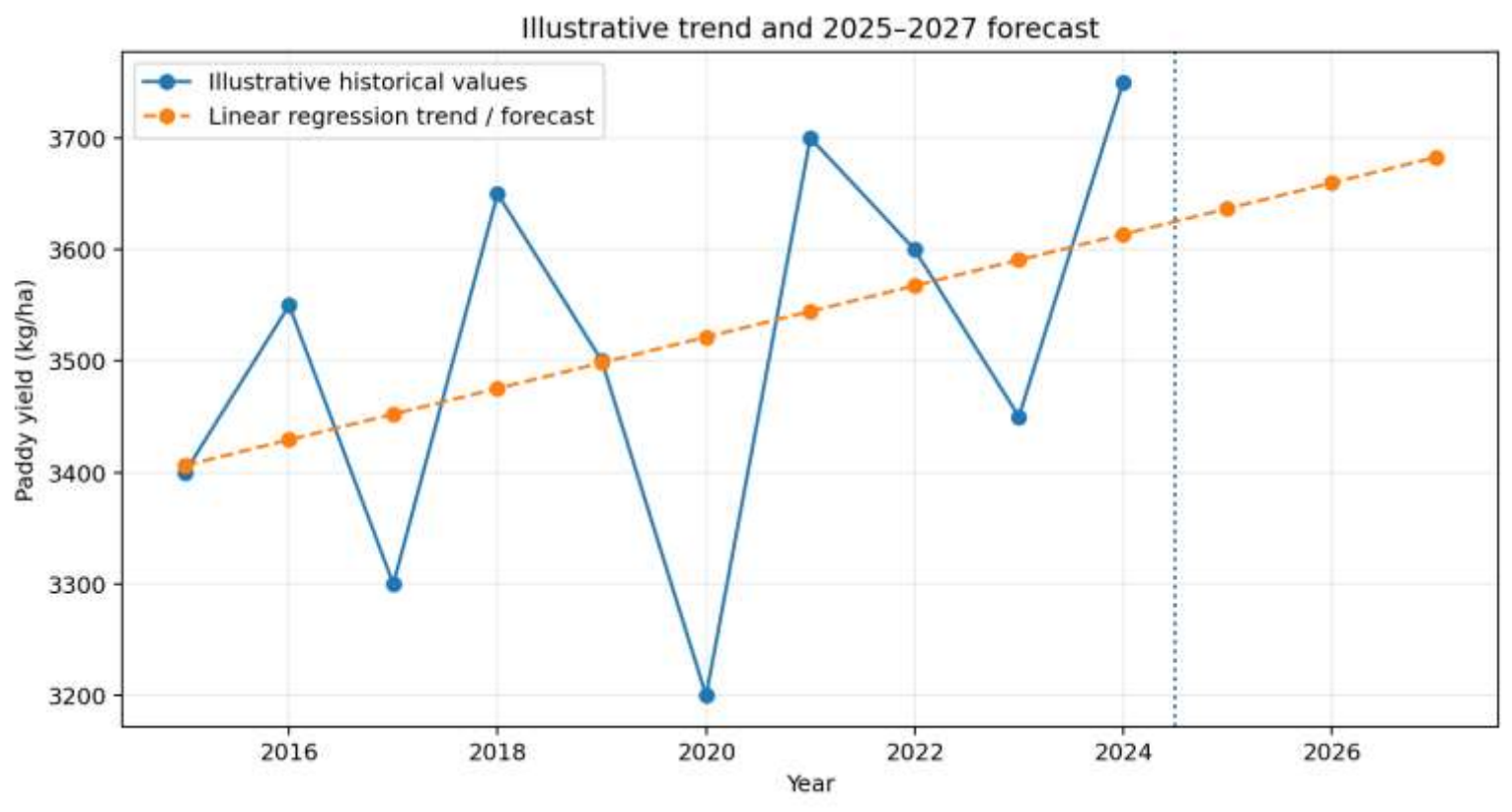
$R^2 = 0.16$

This describes the fit on the illustrative data only.

Use real records before drawing conclusions about Tamil Nadu.

Future yield estimates

A straight-line extension gives a simple baseline, not a guaranteed outcome.



Forecast (illustrative)

2025: 3,637 kg/ha

2026: 3,660 kg/ha

2027: 3,683 kg/ha

These estimates come from the demo line and should not be treated as official forecasts.

Advantages and disadvantages

Regression is useful when we understand what it can—and cannot—tell us.

Advantages

- Easy to understand and explain.
- Shows the direction of a long-term trend.
- Quick to calculate with basic tools.
- Can support early planning and comparisons.

Disadvantages

- Year alone cannot explain weather or farming changes.
- A straight line may miss sudden rises and falls.
- Poor or short data can give misleading estimates.
- Past trends may not continue into future years.

Conclusion

What we learn from the exercise

- Linear regression gives a clear starting point for studying crop-yield trends.
- The best-fit line summarises the direction of the example data and produces year-wise estimates.
- For a dependable Tamil Nadu forecast, official yield records and weather/farm variables are needed.
- The model should be treated as a planning aid, not a promise of the next harvest.